Md Ahsanul Haque, Ph.D.
MAT-120

Quiz-1

Date: July, 2026

Name

DISTRIBUTION OF MARKS		
Question No.	**Full Marks**	**Obtained Marks**
1	10	
2	10	
Total	20	

Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- This assignment contains 2 problems. It is **YOUR responsibility** to check that your copy has all 2 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation and show all logical steps.
- Check that your answer can be read and understood by the grader.

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1. (10 points) Consider the rational function given by $f(x) = \dfrac{kx - 4}{x - 3}$.
   (a) Determine the exact value of the constant k such that the function is its own inverse, meaning $f^{-1}(x) = f(x)$ for all x in the domain.
   (b) Using the value of k found in (i), formally state the domain and range of $f(x)$.

2. (10 points) Solve the following questions
   (a) Let the function
   $$h(x) = \ln\left(x + \sqrt{x^2 + 1}\right)$$
   Demonstrate algebraically that $h(x)$ belongs to the family of odd functions.
   (b) Solve the following equations for x. Be sure to check for any extraneous solutions.
   $$4^x - 3 \cdot 2^{x+1} + 8 = 0$$